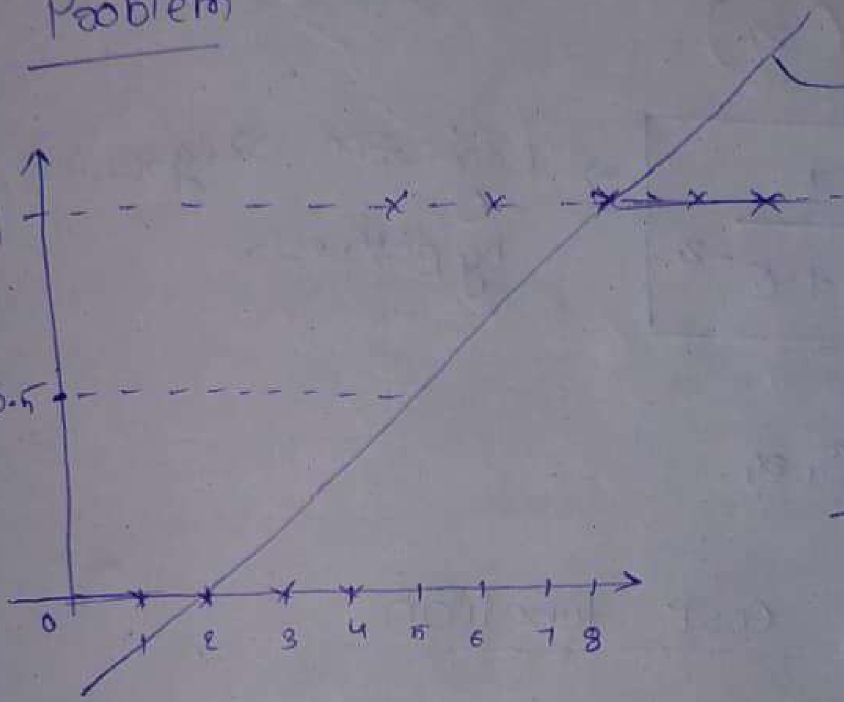
How Logistic Regression solves Classification Problem



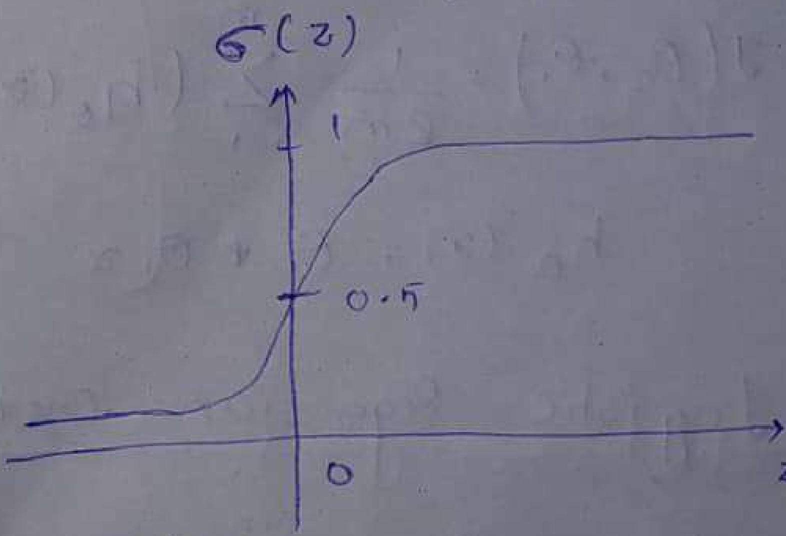
$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1}$$

$\Downarrow$

Best fit line

Sigmoid Activation

$$\frac{1}{1+e^{-z}}$$



$z \geq 0$

$\sigma(z) \geq 0.5$

$$z = \theta_0 + \theta_1 x_1$$

$$h_\theta(x) = \sigma(\theta_0 + \theta_1 x_1) \qquad \sigma = \frac{1}{1+e^{-z}}$$

$$= \sigma(z)$$

$$\boxed{h_\theta(x) = \frac{1}{1+e^{-z}}} \Rightarrow \text{Logistic Regression hypothesis}$$

$$z = \theta_0 + \theta_1 x_1$$

Linear Regression cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)}\right)^2$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

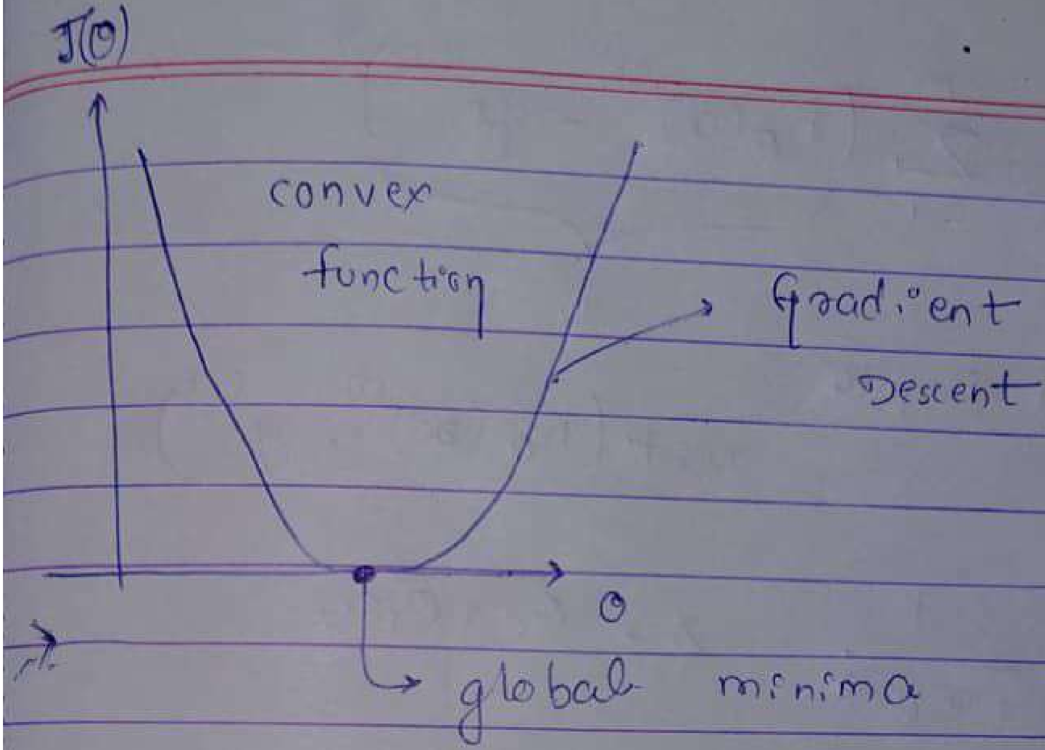
convex function

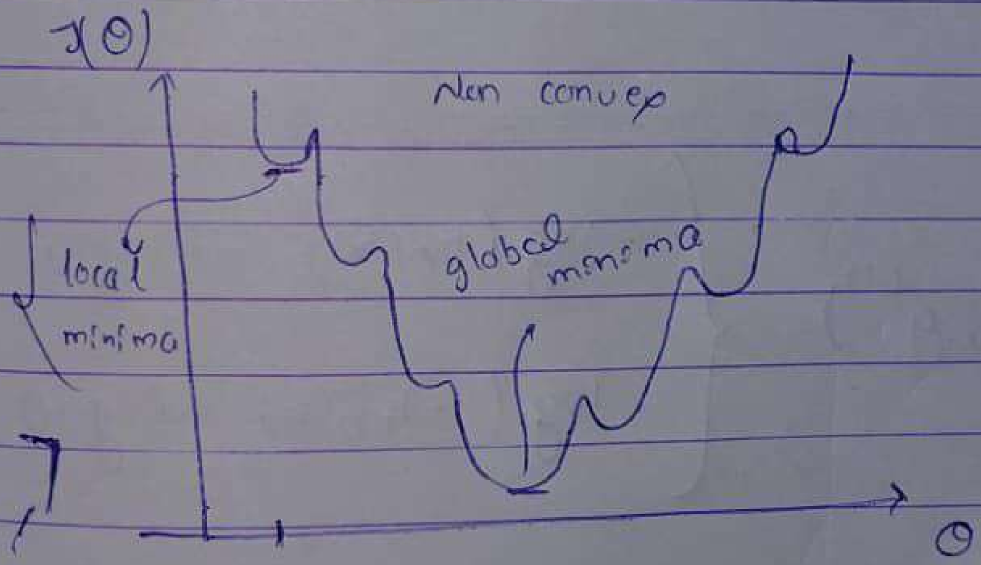
Logistic Regression cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)}\right)^2$$

sigmoid $h_\theta(x) = \frac{1}{1+e^{-z}}$ $\quad z = \theta_0 + \theta_1 x_1$

Non convex function


J(Θ)
convex
function
Gradient
Descent
Θ
global minima


J(Θ)
Non convex
local
minima
global
minima
Θ

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \underbrace{\left(h_\theta(x)^{(i)} - y^{(i)}\right)^2}_{\Downarrow}$$

List Denote

$$\text{cost}\left(h_\theta(x)^{(i)}, y^{(i)}\right)$$

$$h_\theta(x)^{(i)} = \frac{1}{1+e^{-z}} \qquad z = \theta_0 + \theta_1 x_1$$

{log loss}

$$\text{cost}\left(h_\theta(x)^{(i)}, y^{(i)}\right) = \begin{cases} -\log\left(h_\theta(x)\right) & \text{if } y = 1 \\ -\log\left(1 - h_\theta(x)\right) & \text{if } y = 0 \end{cases}$$

$$\text{cost}\left(h_\theta(x)^{(i)}, y^{(i)}\right) = -y \log\left(h_\theta(x)\right) - (1-y)\log\left(1 - h_\theta(x)\right)$$

is

convex function

$$\boxed{J(\theta_0, \theta_1) = \frac{-1}{2m} \sum_{i=1}^{m} \left(y^{(i)} \log (h_\theta(x)^{(i)}) - (1-y^{(i)}) \log (1-h_\theta(x)^{(i)}) \right)}$$

minimize cost function $J(\theta_0, \theta_1)$ by changing θ_0 & θ_1

Convergence Algorithm

Repeat

$\{$ $\quad j = 0$ and 1

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

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